Free balancing equations worksheet worksheet 21

Maths | 8th grade | Algebra

1. Solve for x: 3x + 7 = 2x + 12

A) x = 5
B) x = -5
C) x = 1
D) x = 2

2. If 4(y + 2) = 3y + 10, what is the value of y?

A) y = 2
B) y = 3
C) y = 4
D) y = 5

3. What is the solution to 5 - 2x = 11?

A) x = -3
B) x = 3
C) x = -5
D) x = 5

4. What value of k makes the equation 3k - 9 = 0 true?

A) k = 0
B) k = 1
C) k = 2
D) k = 3

5. If 6m - 2 = 4m + 8, what is the value of m?

A) m = 5
B) m = -5
C) m = 10
D) m = -10

6. Solve for n: 7n + 6 = 2n + 16.

A) n = 0
B) n = -2
C) n = 1
D) n = 2

7. The equation 9p - 3 = 3(p + 1) is given. What is the value of p?

A) p = 1
B) p = 0
C) p = 3
D) p = -3

8. What is the solution to 8 = 2(q - 3) + q?

A) q = 2
B) q = 4

C) q = 3
D) q = 5

9. For which value of x is x/2 + 7 = 10?

A) x = 6
B) x = 5
C) x = 3
D) x = 2

10. Solve for z: 5z - 2(z - 3) = 4 + z.

A) z = 0
B) z = 1
C) z = 3
D) z = 2

Answer Key

1. A
Subtract 2x from both sides to get x + 7 = 12, then subtract 7 to find x = 5.

2. B
Distribute to get 4y + 8 = 3y + 10. Subtract 3y from both sides and then subtract 8 to solve y = 2.

3. A
Subtract 5 from both sides to get -2x = 6, then divide by -2 to find x = -3.

4. D
Add 9 to both sides to get 3k = 9, then divide by 3 to find k = 3.

5. A
Subtract 4m from both sides to get 2m - 2 = 8, then add 2 to both sides and divide by 2 to find m = 5.

6. D
Subtract 2n from both sides to find 5n + 6 = 16, then subtract 6 and divide by 5 to find n = 2.

7. B
Distribute on the right to get 9p - 3 = 3p + 3. Subtract 3p from both sides, then add 3, finally divide by 6 to find p = 0.

8. C
Distribute on the right to get 8 = 2q - 6 + q, simplify to 3q - 6 = 8, add 6 and divide by 3 to find q = 3.

9. A
Subtract 7 from both sides to get x/2 = 3, then multiply by 2 to find x = 6.

10. D
Distribute to find 5z - 2z + 6 = 4 + z, simplify to 3z = -2 + z, then subtract z and divide to find z = 2.